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Team name: Virgo Analytics

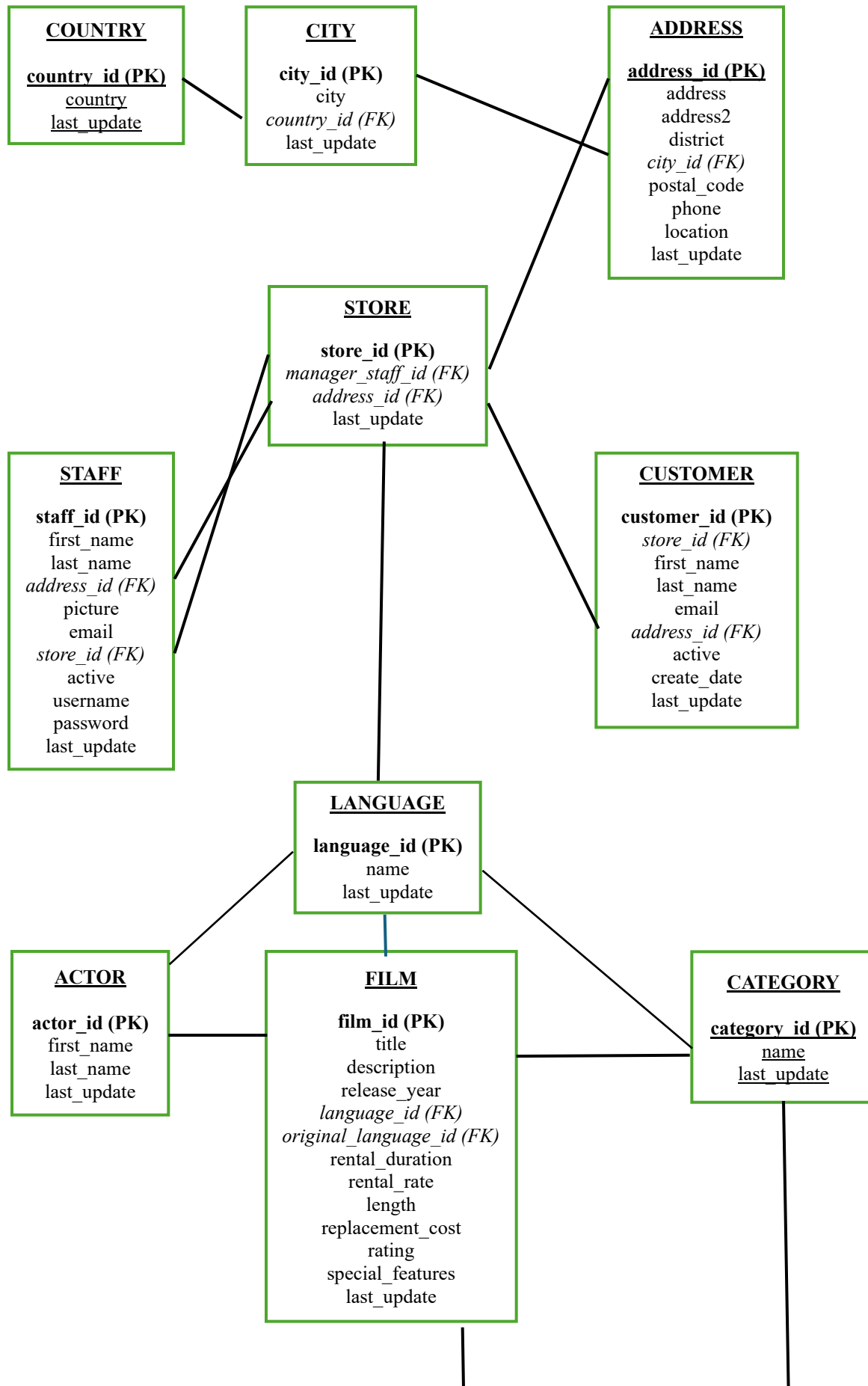
Task 2 - Analyze and Reflect

The Sakila database contains 23 tables and views. The main base tables include actor, film, customer, rental, payment, inventory, language, category, city, country, staff, and store. Each table has a primary key, usually an ID column like actor_id, film_id, customer_id, rental_id, and payment_id.

2. Relationship Observations - Many tables contain foreign keys that connect them to other tables. For example, rental connects to customer, staff, and inventory. Payment connects to customer and rental. Inventory connects to film and store. The junction tables that resolve many-to-many relationships are film actor and film category.

3. Data Size Observations - The tables with the largest number of rows are rental and payment. These are transactional tables because they store ongoing business activity. The smallest tables are sales by store, staff, staff list, and store, each with only 2 rows. The transactional tables include rental, payment, and inventory. The reference tables include language, category, country, and city.

Task 3 GED



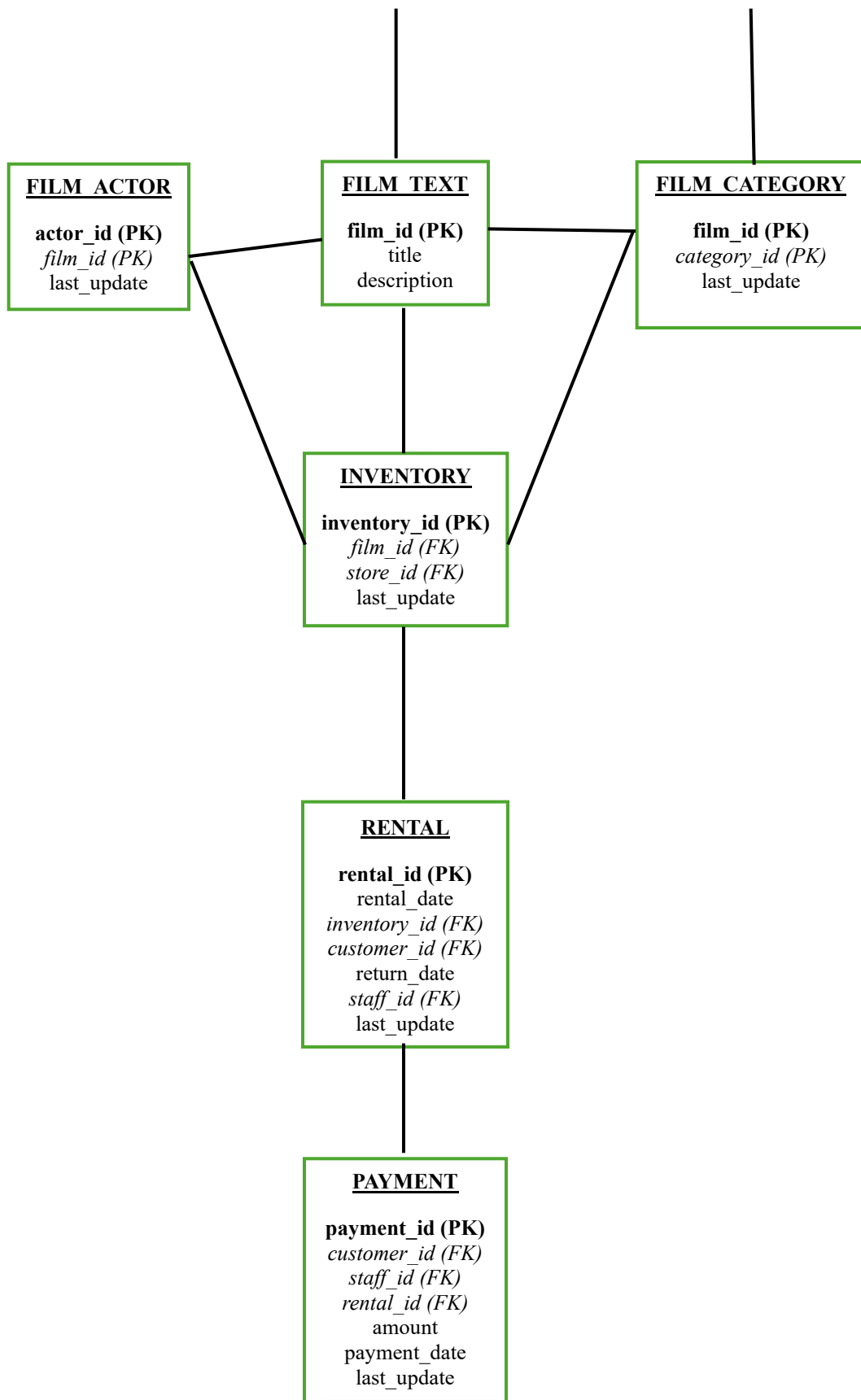


Table Shorthands:

COUNTRY - (country_id PK, country, last_update)

CITY - (city_id PK, city, country_id FK, last_update)

ADDRESS - (address_id PK, address, address2, district, city_id FK, postal_code, phone, location, last_update)

STORE - (store_id PK, manager_staff_id FK, address_id FK, last_update)

STAFF - (staff_id PK, first_name, last_name, address_id FK, picture, email, store_id FK, active, username, password, last_update)

CUSTOMER - (customer_id PK, store_id FK, first_name, last_name, email, address_id FK, active, create_date, last_update)

LANGUAGE - (language_id PK, name, last_update)

ACTOR - (actor_id PK, first_name, last_name, last_update)

FILM - (film_id PK, title, description, release_year, language_id FK, original_language_id FK, rental_duration, rental_rate, length, replacement_cost, rating, special_features, last_update)

CATEGORY - (category_id PK, name, last_update)

FILM_ACTOR - (actor_id PK/FK, film_id PK/FK, last_update)

FILM_TEXT - (film_id PK, title, description)

FILM_CATEGORY - (film_id PK/FK, category_id PK/FK, last_update)

INVENTORY - (inventory_id PK, film_id FK, store_id FK, last_update)

RENTAL - (rental_id PK, rental_date FK, inventory_id FK, customer_id FK, return_date, staff_id FK, last_update)

PAYMENT - (payment_id PK, customer_id FK, staff_id FK, rental_id FK, amount, payment_date, last_update)

Relationship Shorthands:

COUNTRY → CITY [1:M]

CITY → ADDRESS [1:M]

ADDRESS → STORE [1:M]

ADDRESS → STAFF [1:M]

ADDRESS → CUSTOMER [1:M]

STORE → STAFF [1:M]

STORE → CUSTOMER [1:M]

LANGUAGE → FILM [1:M]

ACTOR → FILM_ACTOR [1:M]

FILM → FILM_ACTOR [1:M]

FILM → FILM_TEXT [1:1]

CATEGORY → FILM_CATEGORY [1:M]

FILM → FILM_CATEGORY [1:M]

FILM → INVENTORY [1:M]

STORE → INVENTORY [1:M]

INVENTORY → RENTAL [1:M]

CUSTOMER → RENTAL [1:M]

STAFF → RENTAL [1:M]

RENTAL → PAYMENT [1:M]

CUSTOMER → PAYMENT [1:M]

STAFF → PAYMENT [1:M]

Normalization (3NF)

COUNTRY - 3NF Compliant: Yes - Justification: Country is in 3NF because the country name depends only on country_id is the primary key. Nothing repeats and everything depends on primary key.

CITY - 3NF Compliant: Yes - Justification: City is 3NF because each city depends on city_id, and country_id is just a foreign key linking it to country. Nothing else depends on a non-key field.

ADDRESS - 3NF Compliant: Yes - Justification: Address is in 3NF because all the fields (address, district, postal code, phone, etc.) depend only on address_id. City_id is used as a foreign key.

STORE - 3NF Compliant: Yes - Justification: Store follows 3NF because everything in the table depends on store_id. The manager and address fields are foreign keys and do not create any extra dependencies.

STAFF - 3NF Compliant: Yes - Justification: Staff is in 3NF since all staff details depend on staff_id. The store and address fields are foreign keys and don't cause any transitive dependency.

CUSTOMER - 3NF Compliant: Yes - Justification: Customer is 3NF because all customer information depends only on customer_id. Store and address are just foreign keys and are managed correctly.

LANGUAGE - 3NF Compliant: Yes - Justification: Language is simple and in 3NF because the name depends only on language_id.

Actor - 3NF Compliant: Yes - Justification: Actor is in 3NF because the actor's first and last name depend only on actor_id (primary key). Nothing repeats and everything depends on primary key.

FILM - 3NF Compliant: Yes - Justification: Film is in 3NF because all movie details depend on film_id. Language fields are foreign keys and don't create any dependency problems.

CATEGORY - 3NF Compliant: Yes - Justification: Category follows 3NF since the category name depends only on category_id.

FILM_ACTOR - 3NF Compliant: Yes - Justification: This is a junction table, and it's in 3NF because it only contains the keys that connect actor and film.

FILM_TEXT - 3NF Compliant: Yes - Justification: Film_text is 3NF because the title and description depend directly on film_id, which is the primary key.

FILM_CATEGORY - 3NF Compliant: Yes - Justification: This is also a junction table and is in 3NF because it only connects film and category using their keys.

INVENTORY - 3NF Compliant: Yes - Justification: Inventory is in 3NF because all data depends on inventory_id. Film and store are foreign keys.

RENTAL: 3NF Compliant: Yes - Justification: Rental is 3NF because each rental record depends on rental_id. The foreign keys link it correctly to inventory, customer, and staff.

PAYMENT — 3NF Compliant: Yes - Justification: Payment is in 3NF because all payment information depends on payment_id. The foreign keys are used correctly.